Linaro Connect
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Linaro Automation Appliance preview

Rémi Duraffort
Sahaj Sarup

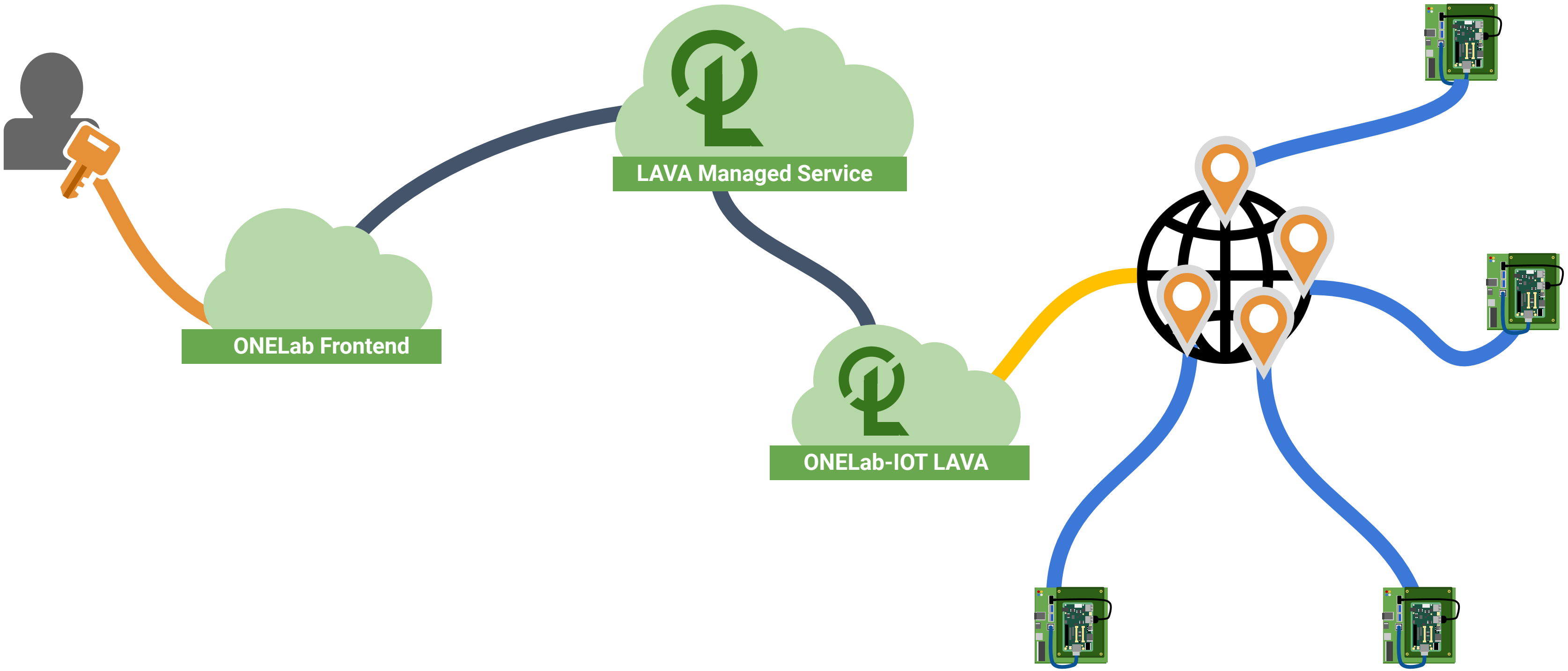
Who we are?

- Rémi Duraffort
  - Principal Tech Lead at Linaro
  - remi.duraffort@linaro.org

- Sahaj Sarup
  - Engineer at Linaro
  - sahaj.sarup@linaro.org

ONELab architecture

ONELab architecture



LAVA

Linaro Automation Validation Achitecture

LAVA brief introduction

- **L**inaro **A**utomated **V**alidation **A**rchitecture
  - **Created** and **maintained** by **Linaro** since **2012**
- Test execution system: **testing software** on **real hardware**
  - **Deploy**, **Boot** and **Test**
- Usages
  - System level testing: **LKFT**
  - Bootloader/firmware testing
  - **ONELab**

- Supports 387 device-types
- Linaro Lab at Cambridge
  - ~ 200 boards

LAVA
linaro.org/lava

Without LAVA

kernel | dtb | rootfs

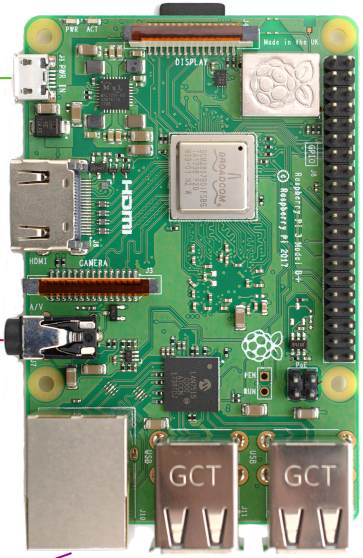
zsh % _

% power on board
% telnet localhost 2000
<enter>
=> dhcp
=> setenv serverip 10.3.1.1
=> […]
=> bootm 0x01000000 - 0x03f00000
[…]
raspberrypi3 login: root
run-test.sh
[…]
% power off board

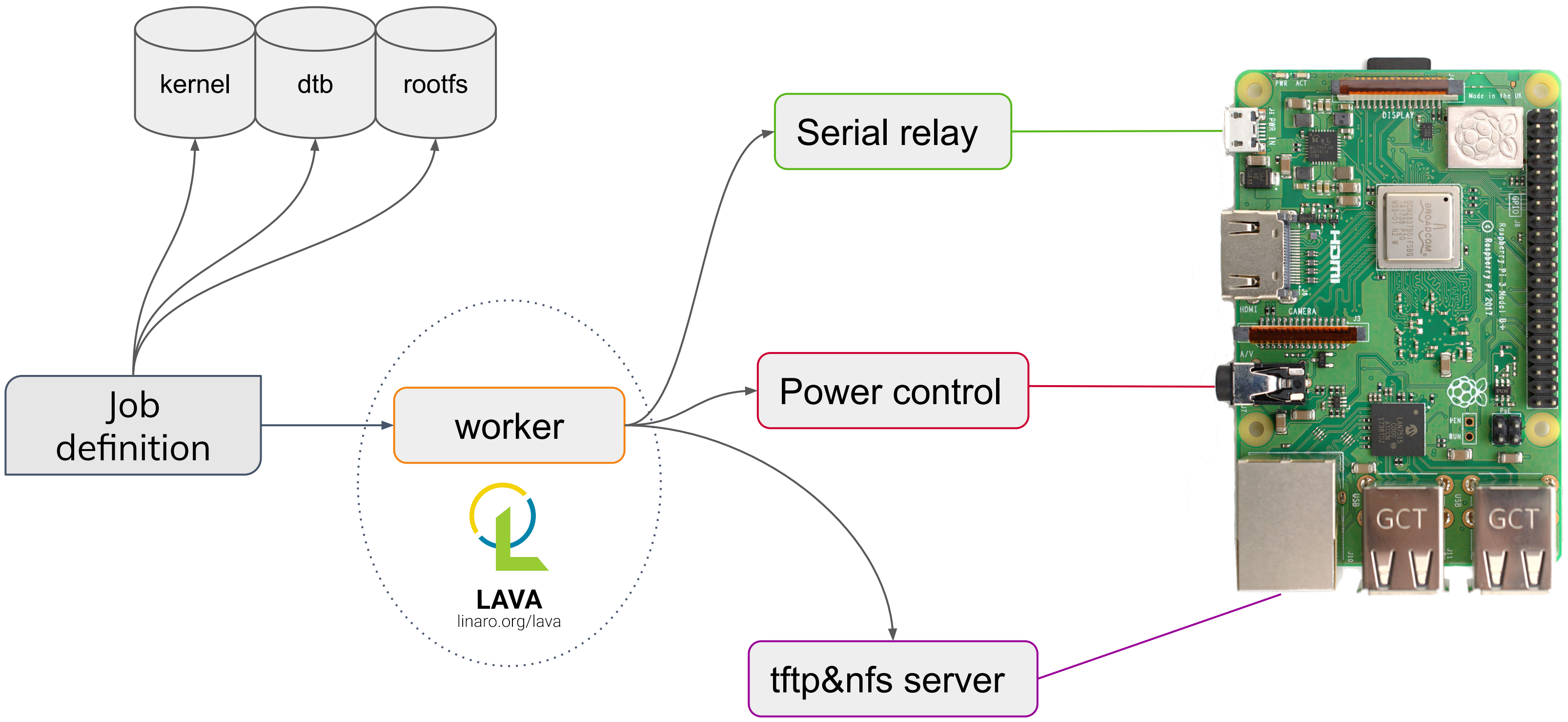
Serial relay

Power control

tftp&nfs server

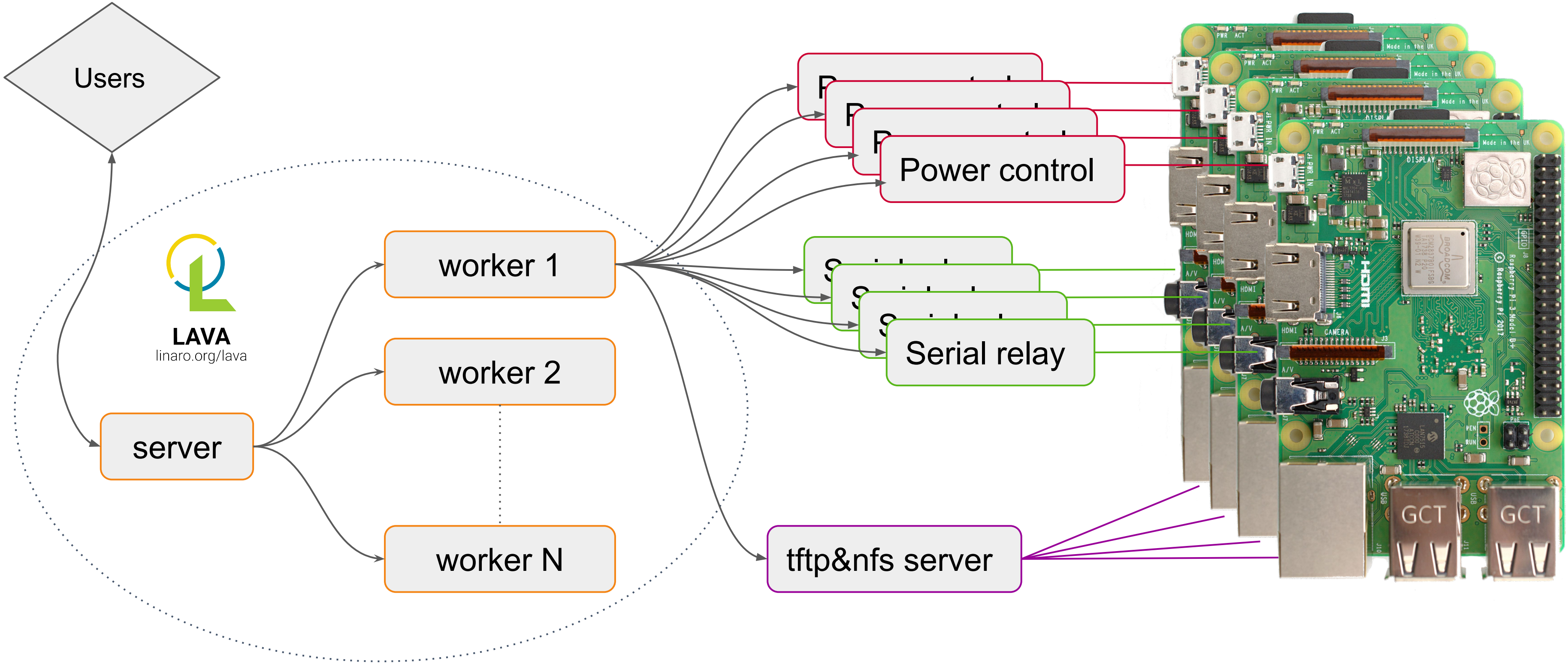


With LAVA



kernel
dtb
rootfs
Job definition
worker
LAVA
linaro.org/lava
Serial relay
Power control
tftp&nfs server

LAVA architecture



LAVA roles

Server

- Web interface/API
  - Submit jobs
  - Results, logs, ...
- Access control
  - Users, Groups, ...
- Scheduling jobs
  - Priority
  - Tags
  - Multinode jobs
- Notifications

- In the **cloud**

Workers

- Deploy resources
- Power On/Off DUTs
- Parse logs
  - Bootloader errors
  - Kernel panic/warnings/OOPS
- Classify errors
  - Infrastructure, Bug
  - Job, Test, ...

- In **private labs**

Supported device-types: 387

aaeon-UPN-EHLX4RE-A10-0864 acer-cb317-1h-c3z6-dedede acer-cbv514-1h-34uz-brya acer-chromebox-cxi4-puff acer-chromebox-cxi5-brask acer-cp514-2h-1130g7-volteer acer-cp514-2h-1160g7-volteer
acer-cp514-3wh-r0qs-guybrush acer-n20q11-r856ltn-p1s2-nissa acer-R721T-grunt adb-nuc alpine-db am335x-sancloud-bbe am437x-idk-evm am57xx-beagle-x15 am6 apq8016-sbc-uboot ar9331-dpt-module arduino101
arduino-nano-33-ble armada-370-db armada-370-rd armada-3720-db armada-3720-espressobin armada-375-db armada-385-db-ap armada-388-clearfog armada-388-clearfog-pro armada-388-gp armada-398-db
armada-7040-db armada-8040-db armada-xp-db armada-xp-gp armada-xp-linksys-mamba armada-xp-openblocks-ax3-4 arndale asus-C433TA-AJ0005-rammus asus-C436FA-Flip-hatch asus-C523NA-A20057-coral
asus-CM1400CXA-dalboz asus-cx9400-volteer at91rm9200ek at91sam9261ek at91sam9g20ek at91sam9m10g45ek at91sam9x25ek at91sam9x35ek at91-sama5d2_xplained at91-sama5d4_xplained ava avenger96 avh
b2120h410 b2260 bcm2711-rpi-4-b bcm2835-rpi-b-rev2 bcm2836-rpi-2-b bcm2837-rpi-3-b-32 bcm2837-rpi-3-b beaglebone-black-barebox beaglebone-black beagle-xm b-u585i-iot02a cc13x2-launchpad cc3220SF cubietruck
cy8ckit-064s0s2-4343w d02 d03 d2500cc da850-lcdk de0-nano-soc dell-latitude-3445-7520c-skyrim dell-latitude-5300-8145U-arcada dell-latitude-5400-4305U-sarien dell-latitude-5400-8665U-sarien disco-l475-iot1 docker
dove-cubox dra7-evm dragonboard-410c dragonboard-820c dragonboard-845c e850-96 frdm-k64f frdm-kw41z fsl-ls1012a-rdb fsl-ls1028a-rdb fsl-ls1043a-rdb fsl-ls1046a-frwy fsl-ls1046a-rdb fsl-ls1088a-rdb fsl-ls2088a-rdb
fsl-lx2160a-rdb fsl-lx2162a-qds fsl-s32v234sbc fvp hi6220-hikey-bl hi6220-hikey hi6220-hikey-r2 hi960-hikey hifive-unleashed-a00 hifive-unmatched-a00 highbank hip07-d05 hp-11A-G6-EE-grunt hp-14b-na0052xx-zork
hp-14-db0003na-grunt hp-x360-12b-ca0010nr-n4020-octopus hp-x360-12b-ca0500na-n4000-octopus hp-x360-14a-cb0001xx-zork hp-x360-14-G1-sona hsdk i945gsex-qs ifc6410 imx23-olinuxino imx27-phytec-phycard-s-rdk
imx28-duckbill imx53-qsrb imx6dl-riotboard imx6dl-sabreauto imx6dl-sabresd imx6dl-udoo imx6q-nitrogen6x imx6qp-sabreauto imx6qp-sabresd imx6qp-wandboard-revd1 imx6q-sabreauto imx6q-sabrelite imx6q-sabresd
imx6q-udoo imx6q-var-dt6customboard imx6sl-evk imx6sll-evk imx6sx-sdb imx6ul-14x14-evk imx6ull-14x14-evk imx6ull-evk imx6ul-pico-hobbit imx6ulz-14x14-evk imx6ulz-lite-evk imx7d-sdb imx7s-warp imx7ulp-evk
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kirkwood-openblocks_a7 kontron-bl-imx8mm kontron-kbox-a-230-ls kontron-kswitch-d10-mmt-6g-2gs kontron-kswitch-d10-mmt-8g kontron-pitx-imx8m kontron-sl28-var3-ads2 kv260 kvm lava-slave-docker
lenovo-hr330a-7x33cto1ww-emag lenovo-TPad-C13-Yoga-zork lpcxpresso55s69 ls1021a-twr lxc mediatek-8173 meson8b-ec100 meson8b-odroidc1 meson-axg-s400 meson-g12a-sei510 meson-g12a-u200 meson-g12a-x96-max
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meson-sm1-sei610 mimxrt1050_evk minnowboard-max-E3825 minnowboard-turbot-E3826 moonshot-m400 morello mps mt8173-elm-hana mt8183-kukui-jacuzzi-juniper-sku16 mt8186-corsola-steelix-sku131072
mt8192-asurada-rev1 mt8192-asurada-spherion-r0 mt8195-cherry-tomato-r2 musca-a musca-b musca musca-s mustang-grub-efi mustang mustang-uefi n1sdp nexus10 nexus4 nexus5x nexus9 nrf52-nitrogen nucleo-l476rg
nxp-ls2088 odroid-n2 odroid-x2 odroid-xu3 orion5x-rd88f5182-nas overdrive ox820-cloudengines-pogoplug-series-3 panda pc-k10n78 peach-pi pixel poplar qcom-qdf2400 qcs404-evb-1k qcs404-evb-4k qemu-aarch64 qemu
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seco-b68 seco-c61 sharkl2 sm8150-mtp sm8250-mtp sm8350-hdk sm8350-mtp snow soca9 socfpga-cyclone5-socrates ssh stm32-carbon stm32l562e-dk stm32mp157a-dhcor-avenger96 stm32mp157c-dk2
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sun50i-h6-orangepi-one-plus sun50i-h6-pine-h64 sun50i-h6-pine-h64-model-b sun5i-a13-olinuxino-micro sun5i-gr8-chip-pro sun5i-r8-chip sun6i-a31-app4-evb1 sun7i-a20-cubieboard2 sun7i-a20-olinuxino-lime2
sun7i-a20-olinuxino-micro sun8i-a23-evb sun8i-a33-olinuxino sun8i-a33-sinlinx-sina33 sun8i-a83t-allwinner-h8homlet-v2 sun8i-a83t-bananapi-m3 sun8i-h2-plus-bananapi-m2-zero sun8i-h2-plus-libretech-all-h3-cc
sun8i-h2-plus-orangepi-r1 sun8i-h2-plus-orangepi-zero sun8i-h3-bananapi-m2-plus sun8i-h3-libretech-all-h3-cc sun8i-h3-orangepi-pc sun8i-r40-bananapi-m2-ultra sun9i-a80-cubieboard4 synquacer-acpi synquacer-dtb
synquacer synquacer-uboot tc2 tegra124-nyan-big thunderx2 thunderx upsquare vexpress x15-bl x15 x86-atom330 x86-celeron x86 x86-pentium4 x86-x5-z8350 xilinx-zcu102

LAVA SW and HW

- **Software** automation
  - LAVA the **de-facto standard**
  - Stable, maintained and well-known

- **Hardware** automation
  - Not any standard
    - Only **custom** solutions
  - Multiple component and options
    - FTDI
    - Power relay/PDU
    - Worker hardware

Linaro Automation Appliance

Demo

Custom setup vs LAA

Custom setup vs LAA

- Custom setup
  - Hardware
    - LAVA worker
    - FTDI cable (for serial)
    - Remotely controllable relay
    - USB cable (to cut)
  - Software
    - Install a LAVA worker (debian)
    - Create the worker in LAVA, get the token and configure it
    - Create the device-type
    - Create the device and assign it to the worker
    - Create the device-dictionary
      - Serial port
      - Web relay command to use
      - IP of the worker
- LAA
  - Register the appliance
  - Set the device-type of the DUT

Custom setup vs LAA

- Custom setup
    - Hardware
        - LAVA worker
        - FTDI cable (for serial)
        - Remotely controllable relay
        - USB cable (to cut)
    - Software
        - Install a LAVA worker (debian): lava-worker, nfsd, tftpd, httpd, …
        - Create the worker in LAVA, get the token and configure it
        - Create the device-type
        - Create the device and assign it to the worker
        - Create the device-dictionary
            - Serial port
            - Web relay command to use
            - IP of the worker
- LAA
    - Register the appliance
    - Set the device-type of the DUT

Weeks

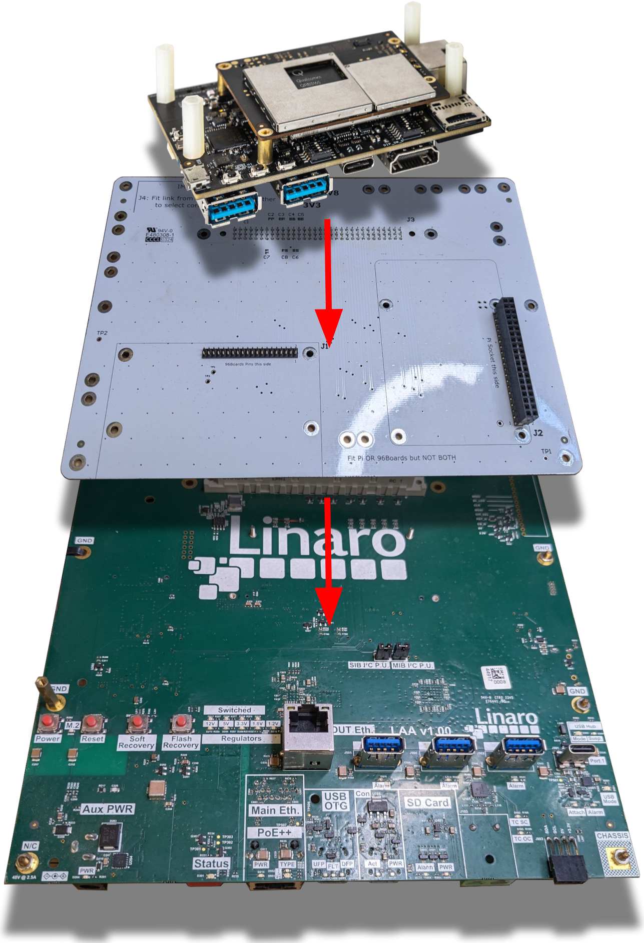
Minutes

Design Goals

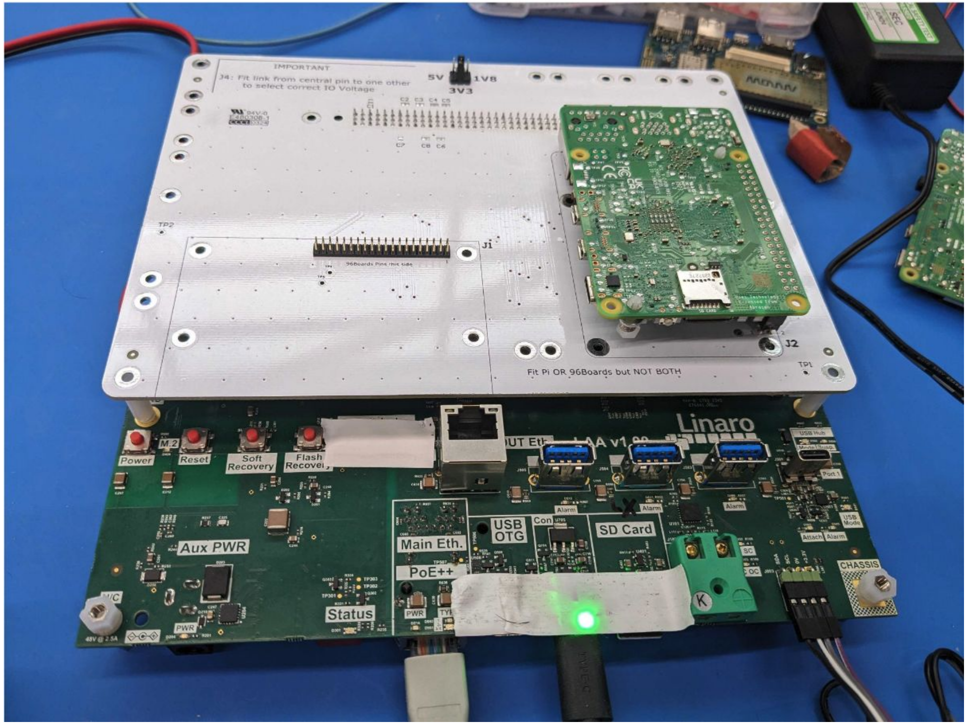
- Drastically reduce cost to host and maintain DUTs
  - Maximize DUT utilization (no idle staging or developer instances)
  - Improve operational and space efficiency through automation harness standardization
  - Reduce testing harness reliability problems
  - Reduce DUT enablement costs
  - Move automation servers out of embedded device labs
- Enroll remote-lab hosted DUTs into CI programs quickly
  - Easy to onboard, reliable, and low cost to maintain for remote-lab partners
- Securely share devices across programs, communities, customers and partners
- Standardize hardware and software automation to reduce testing costs

LAA architecture

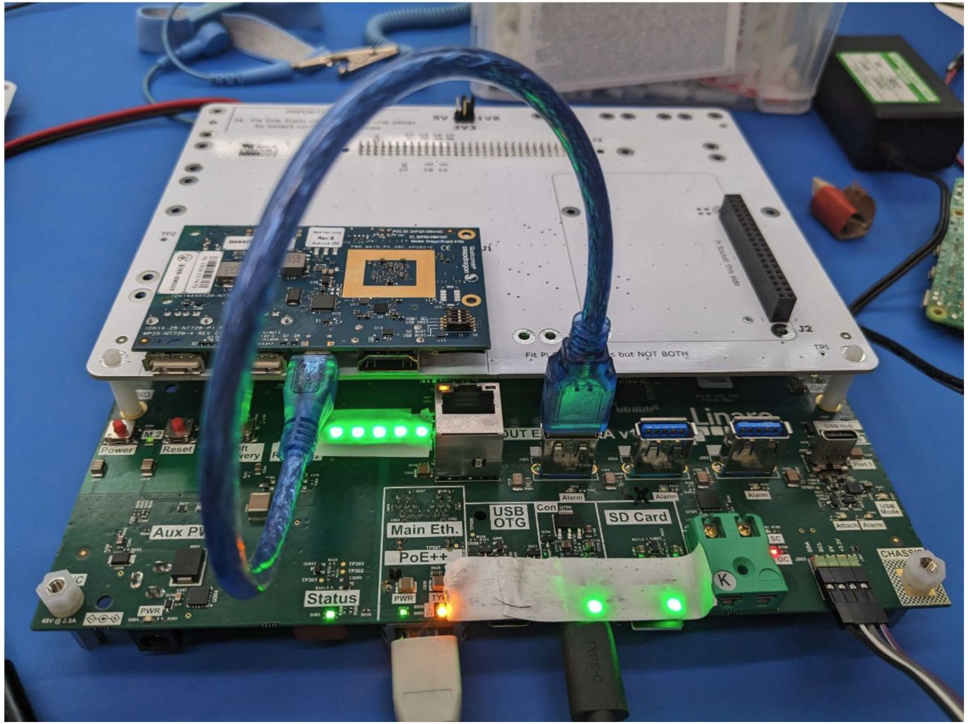
- **L**inaro **A**utomation **A**ppliance
  - **S**tandard **I**nterface **B**oard
  - **M**echanical **I**nterface **B**oard
  - **D**evice **U**nder **T**est
  - LAA=SIB+MIB+DUT
- SIB
  - ARM64 Linux host
  - Standard LAVA worker
    - LAVA auto-configured
- A MIB for each device-type kind
  - Mechanically secure the DUT
  - Use the header (if available) for power, serial, usb, ...



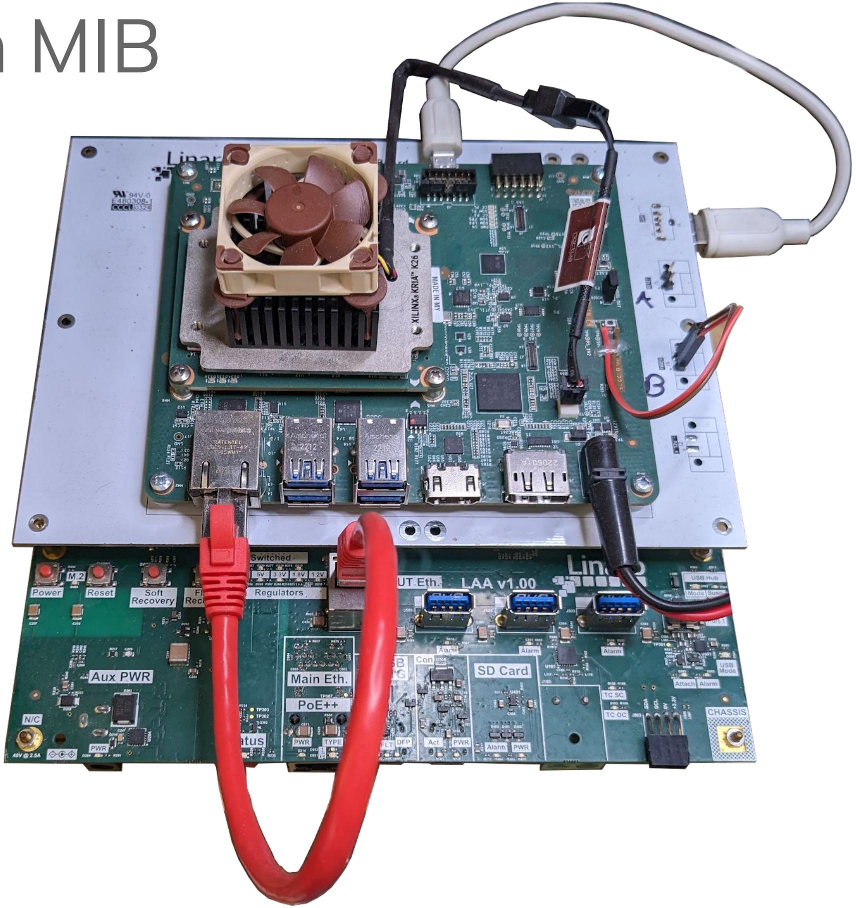
RPi4B on RPi MIB



db410c on 96boards MIB



kv260 on custom MIB



LAA features

- All in one solution
  - Full LAVA worker
    - plug-and-test
  - Automation HW
    - serial, PDU (1v8, 3v3, 5v, 12v), USB, GPIOs, ethernet, relays, thermometers, power consumption, private ethernet, virtual buttons
  - Reproducibility
    - Exact same environment from one DUT to another

- Fleet management
  - Remote upgrades
  - Automatic configuration (auto-connect to LAVA instances, auto-configure attached DUT)

- MIB customization
  - Adapt to each device-type kind

Credits (alphabetical)

- Automation Team
  - Antonio Terceiro
  - Chase Qi
  - Fathi Boudra
  - Paweł Szymaszek
  - Ryan Arnold
  - Stevan Radaković
- Lab Team
  - Elie Hebbo
  - Walid Ghoggali
  - James Howlett
- Linaro people
  - Ilias Apalodimas
  - Sumit Garg
- ...

Questions?

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Thank you

remi.duraffort@linaro.org
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